

GITHUB REPOSITORY: <https://github.com/orange-bannano/django-backend>; Please Ignore comments; STATUS = moving forward reviewing & documenting (postman and commneting) & removing excess fail-safes

Started on	Status	Task	Position	Comments
05/26	Complete	Brief learning on backend concepts and inner workings fo web framework (REST API and its methods (PUT, GET, etc.), MVT, ORM, idempotency, transactions in database, http codes, http communication and headers, jwt, session cookies, refresh tokens, csrf tokens, throttling, rate limiting and other basic security implementations to ensure security, OpenAPI, async operations, filtering and partial loading, security vulnerability like code injections (SQL, JS, etc.))	Refer README.md	
05/27	Complete	Installation of WSL, Python, Pycharm	Refer commands.txt	
05/27	Complete	Installation of XServer (VcXsrv), Xterm, MySQL(server on WSL), Workbench (Windows) and Auth	Refer commands.txt	
05/27	Complete	Django project initialiaiztion	Refer django-init.txt and roadmap.md	
05/28	Complete	Creating 1st project (Mosh's Tutorial)	Creating endpoints with templates, serializations, model creations installing debugging app, creating apps (high cohesion, min coupling)	
05/29	Complete	Static data loading from local & dynamic from db into html (CSS, JS, IMGs)	(CSS, JS, IMGs) TODO: Ginger notation, HTML & CSS (syntax) (https://docs.djangoproject.com/en/6)	
05/30	Complete	Using django shell	TODO: learning commands from off doc (syntax)	
05/30	Complete	JSON parsing	TODO: learning functions and their working in json library on python	
05/31	Complete	Integrating mySQL server running on WSL to Django	Migrated to MySQL from SQLite3, refer django-init.txt	
05/31	Complete	Pandas Basic & ipynb	Basic Pandas , https://www.dataquest.io/blog/jupyter-notebook-tutorial/	
05/31	Complete	Django admin	https://www.youtube.com/watch?v=SlyxjRj8VNY&list=PLsyebzWxl7r2uKvGtqIQcl-	
06/01	Complete	EDA	Exploratory Data Analysis(EDA): Python I Towards Data Science	
06/06	Complete	Authenticate	Creating Users with standard user model. Creating custom authentication_backend classes from Model (default) Installed UI for user	
06/06	Complete	Authoriation	Authorizing based on patterns: ownership, group/roles, is_staff (admin) Groups : Employee, Manager, Admin Employee = can view/modify owned notes, Manager = can view/modify employee ar All operations are to be carried out via notes_collection view using different http me	
06/04	To Do	Create OpenAPI machine readable code	https://chatgpt.com/share/6a2b22d1-faf8-8324-b4b4-4d9f484bd222	
06/07	Complete	request validation	Passing request JSONs throught strict methods, to verify API requests.	
06/12	Complete	Multi query filtering using .getlist and pagination for efficiency		
06/10	Complete	SQL Injections	escaping malicious queries using Django ORM ie queryset.filter(query)	
06/15	Complete	JS Injections	Creating html representation of user's data (malicious) by using render() and creating	
06/12	In Progress	Transactions	if something breack in middle of logic, executed logic can be rolled back, need replicable db? with transactions.atomic() ?	
06/15	Complete	rate limiting	User & IP address based rate limiting through django_ratelimit and custom service via django.core.cache using Redis servered hosted on WSL	
06/03	Complete	Implementing Idempotency	With each CRUD request an idempotency-key http header must be sent, if the featur Each idem key is mapped to a cache key which is mapped to a 'record' dictionary, co If our django server dies mid operation, all request will be stuck in pending status. Need to create a call which create our UUID	
06/19	In Progress	Automated Security auditing of project		
06/05	In Progress	Malicious HOST headers Injection	Whitelisting host domain, preventing malicious/ fake host headers. Password Reset: a request to reset password endpoint with host header of legit wet The reseter create a response HTTP containing secrete key and sents it to victim us	
06/06				